

Assignment_3 计算机原理与位运算

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题目

P1100 : 高低位交换

位运算, 普及-, <https://www.luogu.com.cn/problem/P1100>

```
i = int(input())
bin_i = format(i, '032b')
bin_i_list = list(str(bin_i))
bin_j_list = bin_i_list[16:] + bin_i_list[:16]
bin_j_str = ''.join(bin_j_list)
j = int(bin_j_str, 2)
print(j)
```

代码运行截图 (至少包含有"Accepted")

R299047306 记录详情

评测信息

源代码

编程语言	代码长度	用时	内存
Python 3	183B	198ms	4.33MB

测试点信息

#1 AC 19ms/3.86MB	#2 AC 20ms/3.84MB	#3 AC 20ms/4.07MB	#4 AC 21ms/3.86MB	#5 AC 19ms/3.80MB	#6 AC 19ms/4.24MB	#7 AC 20ms/3.88MB
#8 AC 20ms/4.33MB	#9 AC 21ms/4.13MB	#10 AC 19ms/3.88MB				

pyw2026

所属题目	P1100 高低位交换
评测状态	Accepted
评测分数	100
提交时间	2026-09-24 18:08:16

LC136 : 只出现一次的数字

bit manipulation, Easy, <https://leetcode.cn/problems/single-number/>

此题利用了异或运算的两个关键性质：

- 1. 异或运算具有交换律和结合律
- 2. $0^n=n$; $n^n=0$

```

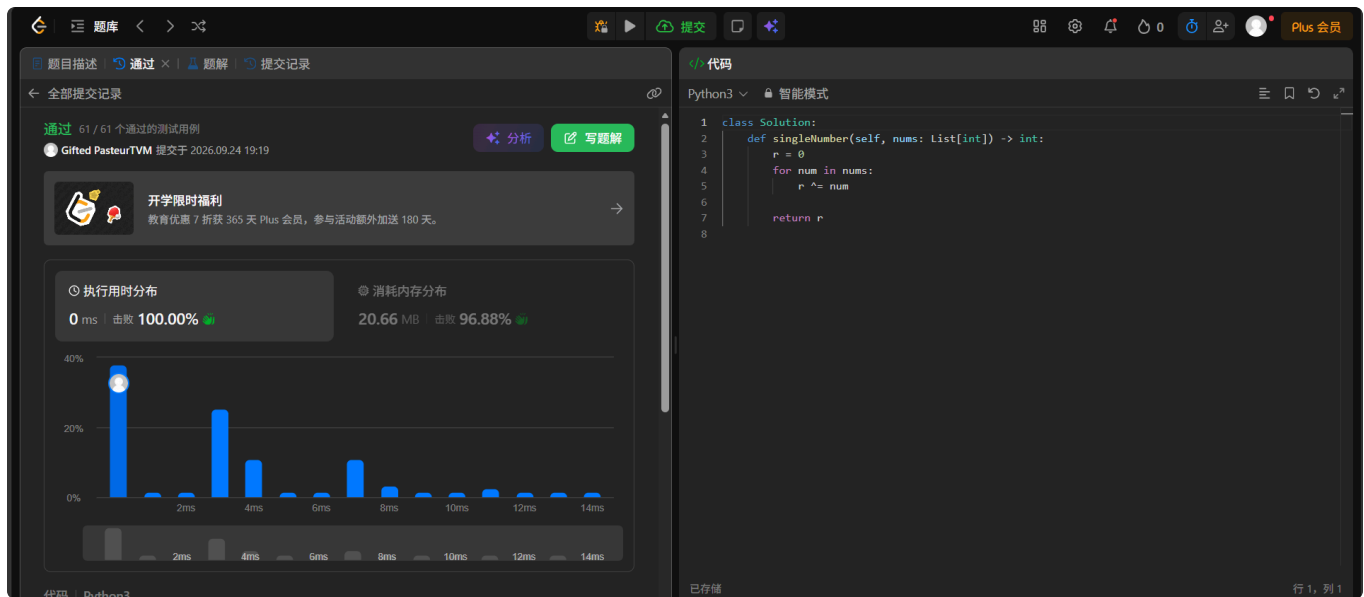
from typing import List

class Solution:
    def singleNumber(self, nums: List[int]) -> int:
        r = 0
        for num in nums:
            r ^= num

        return r

```

代码运行截图 (至少包含有"Accepted")



LC191 : 位1的个数

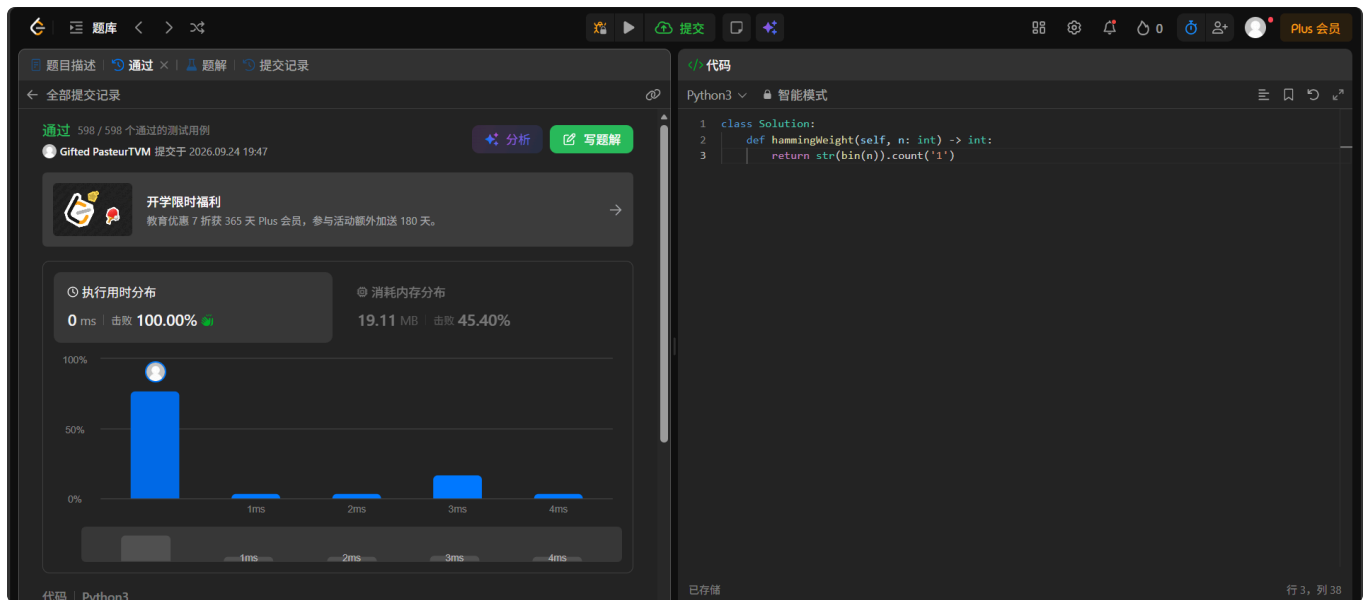
bit manipulation, divide and conquer, Easy, <https://leetcode.cn/problems/number-of-1-bits/>

```

class Solution:
    def hammingWeight(self, n: int) -> int:
        return str(bin(n)).count('1')

```

代码运行截图 (至少包含有"Accepted")



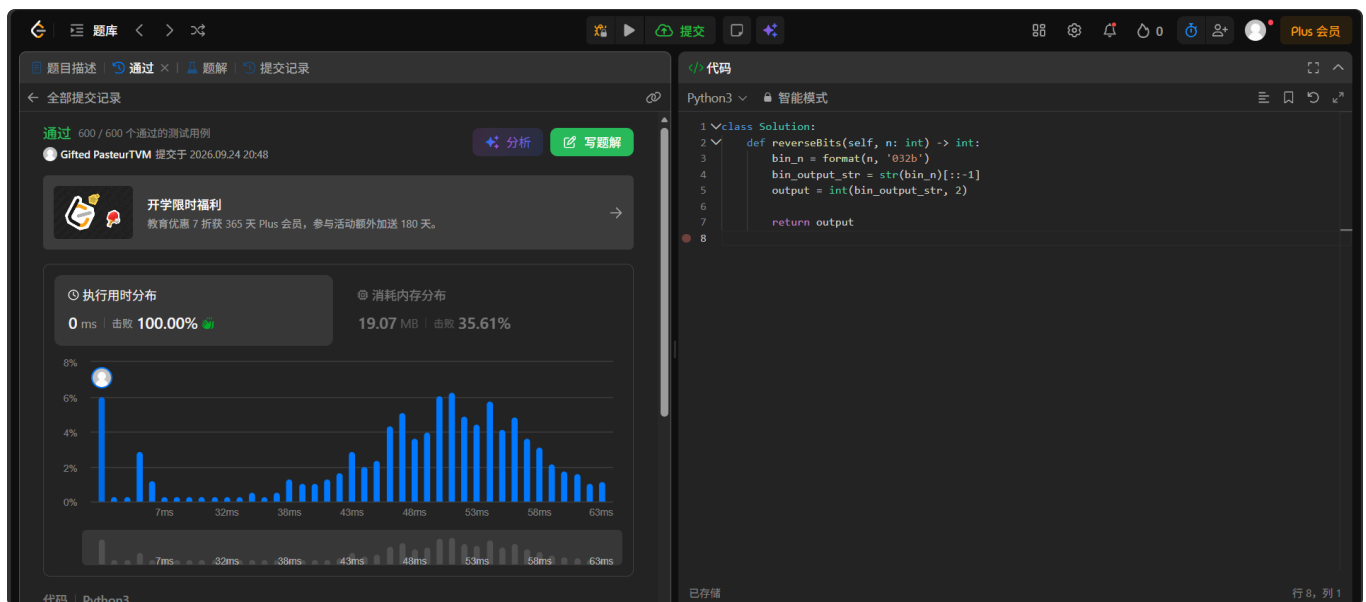
LC190 : 颠倒二进制位

bit manipulation, Easy, <https://leetcode.cn/problems/reverse-bits/>

```
class Solution:
    def reverseBits(self, n: int) -> int:
        bin_n = format(n, '032b')
        bin_output_str = str(bin_n)[::-1]
        output = int(bin_output_str, 2)

        return output
```

代码运行截图 (至少包含有"Accepted")



LC356 : 根据数字二进制下 1 的数目排序

bit manipulation, Easy, <https://leetcode.cn/problems/sort-integers-by-the-number-of-1-bits/>

```
class Solution:
    def sortByBits(self, arr: list[int]) -> list[int]:
        arr.sort()
        arr.sort(key=lambda x: bin(x).count('1'))

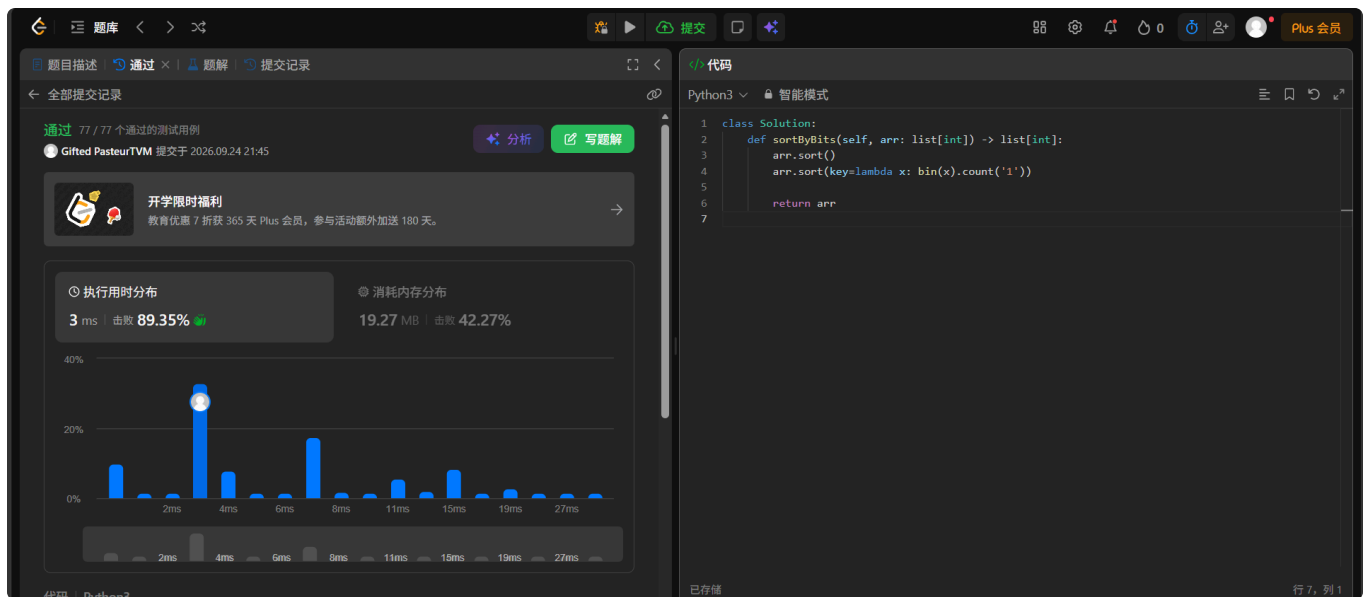
        return arr
```

or

```
class Solution:
    def sortByBits(self, arr: List[int]) -> List[int]:
        l = [(bin(i).count('1'), i) for i in arr]
        l.sort()

        return [n[1] for n in l]
```

代码运行截图 (至少包含有"Accepted")



LC1404 : 将二进制表示减到 1 的步骤数

bit manipulation, Medium, <https://leetcode.cn/problems/number-of-steps-to-reduce-a-number-in-binary-representation-to-one/>

```
class Solution:
    def numSteps(self, s: str) -> int:
        num = int(s, 2)
        steps = 0
```

```
while num != 1:
    steps += 1
    if num % 2 == 0:
        num //= 2
    else:
        num += 1

return steps
```

代码运行截图（至少包含有"Accepted"）

